

CameraIa: An Edge-Side, Browser-Native Framework for Remote Psychophysiological Experiments with rPPG Sensing

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Abstract – CameraIa is an edge-side, browser-native framework for remote psychophysiological experiments with rPPG sensing. It synchronizes task events with face-image-derived feature records and exports local CSV/ZIP files without transmitting or storing raw facial video. An exploratory N=1 author-participant deployment was evaluated as a 10-session, 600-trial browser-based task experiment in laboratory and home settings. The deployment was assessed against a study-specific exploratory 80% behavioral-response reference level and temporally alignable export of task-event and face-image-derived records. The findings support a limited implementation-level conclusion under the tested author-participant, device, environment, posture, lighting, and task-execution context. They do not establish general feasibility, robustness, validity in other settings, rPPG accuracy, or psychological/physiological effects.

Keywords : Browser-Native Sensing, Remote Photoplethysmography, FaceMesh, Edge-Side Processing, Task-Aligned Measurement, Laboratory and Home Settings

1. Introduction

The intended use case of CameraIa is a browser-based remote psychological/psychophysiological task experiment conducted in laboratory and home settings. In this use case, the participant sits in front of a laptop, views task stimuli, and responds by key press. CameraIa is not intended for continuous physiological sensing during ordinary daily activities. In this paper, edge-side processing means that webcam frames are processed within the participant’s browser on the client device, reducing the need to transmit or retain privacy-sensitive raw facial video. In this paper, rPPG sensing refers to acquiring and logging facial ROI color traces for possible subsequent rPPG analysis; the present prototype does not estimate validated heart rate or establish rPPG accuracy. Even in this task-based experiment, deployment outside a controlled laboratory introduces practical difficulties for human-interface research, including heterogeneous illumination, camera placement, posture, browser timing, and local data handling.

Many video-based physiological measurement pipelines rely on raw video recording or server-side processing. Such approaches can be difficult to deploy in private spaces because raw facial video is privacy-sensitive and may be identifiable. An edge-

side browser architecture offers an architectural privacy advantage: raw frames can be processed on the client device, while only extracted features, task events, and quality indicators are stored. This does not constitute a dedicated privacy-preserving algorithm; rather, it is a system-design choice that reduces the need to transmit or retain raw video.

Real-time processing is not treated as the only technically possible architecture. Local video recording followed by offline processing could preserve timestamps if implemented carefully. However, the intended use case of CameraIa is a browser-based remote psychological/psychophysiological task experiment conducted in laboratory and home settings, where retaining identifiable facial video for later analysis can itself be a privacy and acceptability burden. CameraIa therefore extracts face-image-derived records during task execution and stores only task events, exported feature fields, and auxiliary context fields, without raw facial video persistence. In this sense, real-time browser-side processing is used to support feature-only local logging and timestamp alignment within the same client-side runtime, rather than to claim more accurate physiological estimates or online task adaptation.

To address this deployment scenario, we present **CameraIa**, an edge-side, browser-native framework for remote psychophysiological experiments with rPPG sensing.

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1.1 Research Questions

The exploratory Research Questions (RQs) focus on task-synchronized face-derived records and descriptive sensor-derived observations under the tested configuration:

- **RQ1:** In the tested remote psychological/psychophysiological task setting, did Camerale record behavioral responses in at least 80% of completed trials while exporting the task-event and rPPG-sensing records in a temporally alignable form?
- **RQ2:** What kinds of between-environment differences are observable in the physiological signals collected through such a deployment?

A behavioral response was the participant’s F/J key response to the orientation of the presented Gabor stimulus. For RQ1, a completed trial was counted in the numerator when its stored reaction time was between 100 and 1500 ms, inclusive. The RQ1 behavioral-response rate was calculated as the number of such trials divided by the total number of completed trials. Temporally alignable form means that the exported task-event records and face-image-derived records contain timestamps that allow them to be related on the same session timeline. The 80% target was exploratory and study-specific, not preregistered, independently validated, or intended as a general feasibility threshold. We also examined whether the exported records contained face-derived fields, including rPPG-oriented ROI values, motion estimates, landmark-availability summaries, and a frame-to-frame green-channel brightness-change indicator. `mean_quality` is reported only as a descriptive exported-window-level landmark-availability summary; it is not used as the main RQ1 criterion and should not be interpreted as face-detection availability across all camera frames. In RQ2, the term “physiological signals” is used operationally to refer to task-synchronized face-derived and sensor-derived exported records, including rPPG-oriented ROI values and related exported fields. These records are not treated as validated physiological-state measurements. The present design was not able to determine whether laboratory/home differences reflected psychological state, physiological state, lighting, screen light, posture, camera geometry, the measurement pipeline, task timing, participant state, or their mixture. There-

fore, laboratory/home differences were interpreted as deployment-context differences rather than psychological, cognitive, motivational, or physiological effects. These analyses are intended only to describe operational completion and exported-record differences under the tested configuration; they do not demonstrate general deployability across users, devices, or environments.

1.2 Contributions

In this paper, we focus on system integration and conservative deployment observations. The contributions are:

1. **Edge-side browser-native system integration:** An implementation for remote psychological and psychophysiological task experiments that combines task execution, face-derived feature extraction, timestamp alignment, and local persistence within one browser runtime.
2. **Task-synchronized feature logging pipeline:** A logging pipeline that records task events, rPPG-oriented ROI values, FaceMesh-derived motion estimates, exported-record landmark summaries, and frame-derived brightness-change indicators without raw facial video persistence.
3. **Exploratory deployment under limited conditions:** An N=1 author-participant deployment used to assess the study-specific operational criteria under limited laboratory/home conditions.

The remainder of this paper is organized as follows. Section II reviews related work in rPPG, web/edge physiological sensing, and task-measurement platforms. Section III describes the Camerale system architecture. Section IV describes the browser-based task deployment and reporting details. Section V presents descriptive observations. Section VI discusses design requirements and safeguards for future deployments, and Section VII concludes.

2. Related Work

2.1 Remote Physiological Measurement and rPPG

Remote photoplethysmography (rPPG) estimates pulse-related color changes from facial video. Classic approaches include ambient-light remote plethysmographic imaging and blind-source video methods^{[1],[2]}, while more recent work includes deep-

learning-based estimators^[3]. These studies show that camera-based and webcam-based rPPG functionality already exists. They also show that uncontrolled capture conditions, implementation choices, and evaluation pipelines are important factors in remote heart-rate imaging. Accordingly, Cameralea does not claim to introduce a new rPPG estimator or to solve rPPG degradation under uncontrolled conditions. It records rPPG-oriented ROI values and exported indicators so that task-aligned records can be inspected descriptively.

2.2 Browser-Based, Web-Deployable, Edge-Based, and Privacy-Oriented rPPG Systems

Several closely related prior studies and systems are directly relevant to Cameralea’s positioning. Ayesha *et al.* proposed a web application for experimenting with and validating remote measurement of vital signs^[4]. LabVanced provides remote heart-rate detection/rPPG functionality within an online experiment platform^[5]. FacePhys-Demo demonstrates browser-based rPPG monitoring with local inference in a public web demo/repository^[6]. RhythmEdge addresses contactless heart-rate estimation on edge devices^[7]. Gupta and Etemad investigated privacy-preserving remote heart-rate estimation from facial videos^[8]. Di Lernia *et al.* evaluated rPPG via online webcams under uncontrolled conditions^[9]. Together, these related studies and public systems indicate that browser-based, web-deployable, edge-based, and privacy-oriented rPPG have already been explored. Cameralea therefore does not claim novelty in browser-based rPPG functionality, local rPPG processing, edge-oriented heart-rate estimation, or privacy preservation as an algorithm. Instead, Cameralea is positioned as a system-integration implementation for remote psychological and psychophysiological task experiments: it combines browser-based task execution, face-derived feature extraction, task-event/frame-record timestamp alignment, exported-record landmark summaries, and local CSV/ZIP persistence without raw facial video retention within one browser runtime.

2.3 Task Measurement and Physiological Data Collection Platforms

Well-established web experiment libraries and experiment runtimes, such as jsPsych^[10] and PsychoPy/PsychoJS/Pavlovica^[11], support browser-

based task execution and online experiments. However, they do not generally provide an integrated webcam-derived physiological feature pipeline, task-sensing synchronization, and client-side feature logging as a single browser runtime. Conversely, cloud analytics APIs can provide rich video-based analysis but often require transmitting video streams away from the participant’s device. Table 1 maps this design trade-off space.

We structure the initial evaluation as a single-case exploratory deployment. The deployment is not intended to estimate population-level psychological or physiological effects. Instead, it documents whether the integrated browser runtime can complete the planned browser-based task logging procedure and what measurement-quality differences are observed in the tested laboratory and home contexts.

3. System Architecture: Cameralea

Cameralea (derived from "Camera" + "Laboratory") is positioned as an edge-side, browser-native research implementation for remote psychological and psychophysiological task experiments. Its contribution is not browser-based rPPG, local rPPG processing, or privacy preservation as an algorithm. Rather, it integrates task execution, face-derived feature extraction, task-event/frame-record timestamp alignment, exported-record landmark summaries, and local CSV/ZIP persistence without raw facial video retention within one browser runtime. Although the deployment in this paper used one Gabor-orientation task, the runtime separates task logic from the sensing pipeline so that other task procedures can be implemented without changing the feature-extraction loop.

3.1 High-Level Architecture

The system enables a browser-based task experimental loop inside the browser sandbox (Fig. 1). The architecture consists of three coupled modules:

1. **Task Engine:** Stimulus rendering, trial logic, key responses, feedback, staircase updates, and reaction-time logging.
2. **Vision Pipeline:** Webcam ingestion, MediaPipe Face Mesh inference, rPPG-oriented ROI feature extraction, head-motion estimates, blink/EAR features, and frame-level quality indicators.
3. **Data Manager:** Timestamp alignment,

Table 1 Design Trade-off Mapping of Data Collection Platforms

System / Platform	On-device video processing	Task-sensing synchronization	syn-Client-side persistence	per-Raw export	video Typical use
Web experiment libraries	N/A (task only)	Partial	Depends on setup	N/A	Behavioral tasks
Experiment runtimes	Depends on deployment	Yes for task events	Depends on setup	Depends	Lab/online experiments
Browser or web rPPG systems	Often possible	Varies by platform	Varies	Varies	Vital-sign estimation
Cloud sensing APIs	No	Low or asynchronous	No	Often required	Video analytics
Camera	Client-side feature extraction	Task events and frame features one runtime	Local ZIP export of logs	Not used by Camera	Remote task studies

Note: The table summarizes design emphasis rather than categorical superiority. Capabilities depend on implementation and deployment configuration.

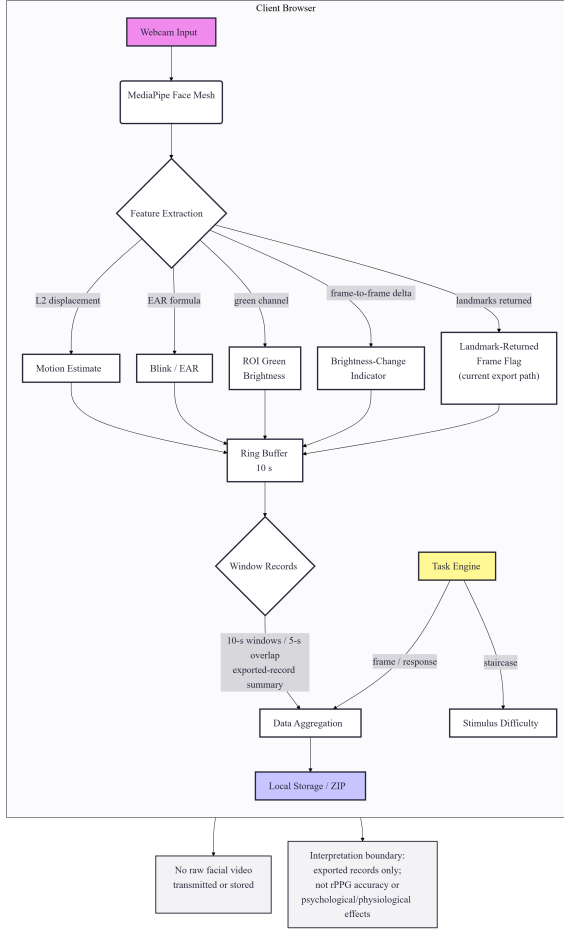


Fig. 1 Overview of the Camera system architecture. The browser obtains webcam frames, extracts face-derived exported fields, aggregates 10-s window records, synchronizes them with task events, and exports local CSV/ZIP logs. Under the tested implementation, only extracted fields and task events are logged; raw facial video is neither transmitted nor stored by Camera.

trial/frame aggregation, CSV serialization, ZIP packaging, and local persistence without raw-video storage.

3.2 Technical Implementation: The Vision Pipeline

The current implementation uses MediaPipe Face Mesh as a software framework for browser-based facial landmark extraction^[12]. The camera stream is requested through `getUserMedia` with ideal constraints of 640×480 pixels and 30 fps. Because actual camera settings can depend on the browser and device, these constraints should be interpreted as requested settings rather than verified hardware settings.

3.2.1 Feature Extraction Logic

The present implementation records sensor-derived features and quality indicators rather than validated clinical or medical measurements. Two categories are central to this paper:

1. Head Motion Index (M_t) Head motion is approximated from the frame-to-frame displacement of a facial landmark. Let $L^t = (x^t, y^t)$ be the normalized landmark coordinate at frame t . The motion scalar is computed as:

$$M_t = \|L^t - L^{t-1}\|_2. \quad (1)$$

This feature is retained as a motion-related exported field for descriptive inspection of task-synchronized records. It is not an algorithmic correction for motion artifacts in rPPG and is not used to compensate for motion-induced rPPG degradation.

2. ROI Green-Channel Value and Brightness-Change Indicator The implementation samples a small forehead region around a FaceMesh landmark and records the green-channel brightness as an rPPG-oriented ROI value. It also computes a frame-to-frame green-channel brightness-change indicator:

$$B_\Delta(t) = |G_t - G_{t-1}|, \quad (2)$$

where G_t is the downsampled frame-level green-

channel average at frame t . Camerale is presented as a deployment and logging architecture, not as a method for improving rPPG signal quality under motion or illumination changes. Instead, it records motion-related and brightness-related exported fields for descriptive inspection of exported records. The brightness-change indicator does not measure camera exposure settings, external illuminance, specific environmental events, psychological or physiological state, or rPPG accuracy.

3.3 Technical Implementation: The Task Engine

The experimental psychophysical probe is a two-alternative forced-choice orientation discrimination task. Using the HTML5 Canvas API, the task engine procedurally renders oriented grating stimuli, records key responses, calculates reaction time with `performance.now()`, and updates task difficulty. Procedural rendering avoids network-dependent stimulus loading during a trial and keeps task timing in the same browser runtime as feature logging.

3.3.1 Timing and Synchronization

The implementation uses the following timing strategy:

1. **High-resolution timestamps:** Task events and frame features are timestamped using `performance.now()`.
2. **Frame-loop alignment:** Feature extraction and task rendering are coordinated with the browser frame loop.
3. **Local runtime alignment:** Task events and sensor-derived features are logged in the same browser runtime, avoiding server-side reconciliation.

3.4 Data Management and Client-Side Persistence

The current implementation uses JSZip to export local log files as a ZIP archive. The archive contains `metadata.json`, `trials.csv`, `features_raw.csv`, `windows.csv`, `blocks.csv`, and `subjective.csv`. The metadata file stores the session ID, start time, user agent, and block plan. The user initiates the download at the end of the session. This design avoids Camerale-side raw-video storage and avoids raw-video transmission, but it also reduces the ability to inspect image artifacts retrospectively. For this reason, Camerale logs reproducible exported fields

Algorithm 1 Camerale Task Loop (Simplified)

```

Initialize Camera, FaceMesh, Canvas
StartContinuousVisionFrameLoop()
while Session Active do
  Block  $\leftarrow$  GetCurrentBlock()
  Context  $\leftarrow$  Block.instruction
  ShowInstruction(Context.text)
  for trial  $i = 1$  to  $n$  do
     $\theta \leftarrow$  SampleOrientation( $\{-45^\circ, +45^\circ\}$ )
    ShowFixation(400–600 ms)
     $t_{onset} \leftarrow$  performance.now()
    DrawOrientationStimulus( $\theta$ , 200 ms)
    while No KeyPress do
      Poll F/J KeyPress
    end while
     $t_{response} \leftarrow$  performance.now()
     $RT \leftarrow t_{response} - t_{onset}$ 
    LogTaskEvent( $RT$ , Accuracy, Context)
    UpdateStaircase(Accuracy, relative  $\pm 5\%$  contrast step)
    LinkTaskEventToFrameTimestamps( $t_{onset}$ ,  $t_{response}$ , Context)
  end for
end while

```

such as rPPG-oriented ROI green-channel values, head-motion estimates, and a frame-to-frame green-channel brightness-change indicator.

4. Methodology

4.1 Target Usage Scenario and Exploratory Deployment

The deployment was a browser-based task experiment conducted in one laboratory setting and one home setting. The participant sat in front of a laptop, viewed Gabor stimuli, and responded using the keyboard. This setting should be distinguished from unconstrained daily-life physiological sensing during walking, conversation, cooking, sleep, or other ordinary activities.

To evaluate operational completion under this scenario, we conducted an exploratory longitudinal deployment with one author-participant. The deployment consisted of 10 sessions and 600 total trials. Each session consisted of six 10-trial blocks, with two blocks for each instructional task context (neutral instruction, negative instruction, and positive instruction), as reflected in the exported block plans. The order of blocks was randomized by the web application.

4.2 Participant (Autoethnography)

The participant was the primary author. This was a self-experimentation evaluation: no other participants were involved, no identifiable raw video was transmitted by Cameralab, and the author provided informed consent for self-participation. Because the participant was also the author and knew that the task instructions were experimentally constructed, the instruction blocks cannot be treated as validated manipulations of evaluation pressure, reward motivation, cognitive state, motivational state, or psychological state. They are used only as exploratory task-interface contexts for demonstrating synchronized logging across task conditions.

4.3 Hardware and Software Configuration

Table 2 summarizes the hardware and software information available for the deployment. Values that were not logged during the deployment are explicitly marked as not logged rather than inferred from the revision-time environment.

4.4 Deployment Environment and Lighting Conditions

The deployment was conducted during daytime in an indoor university laboratory/research room and an indoor private home room under general artificial room lighting. Both rooms had windows, but curtains were closed during the sessions. Direct natural light was not used as the primary illumination source in either context. The built-in laptop display brightness was fixed across sessions. Because the deployment used camera-derived setup checks rather than external light-meter measurements, this paper reports brightness ranges only as author-confirmed setup/check information. The camera-derived brightness range observed during setup/checks was 105–114 in the laboratory setting and 102–111 in the home setting. Table 3 summarizes the deployment environment and lighting conditions.

4.5 Experimental Procedure and Instructional Task Contexts

Each session began with a pre-session capture check for basic camera conditions, including approximate camera-derived brightness and pose, before the camera feed was hidden to reduce distraction. The task included negative, neutral, and positive instruction contexts as exploratory task-interface conditions. These contexts were used to demonstrate that Cameralab can present different instruction frames while recording synchronized behavioral and face-derived records. The labels “negative” and “positive” refer only to instruction labels used in the task interface. Because the participant was the author and knew that the instructions were experimentally constructed, these contexts are not treated as validated manipulations of evaluation pressure, reward motivation, cognitive state, or motivational state.

4.6 Psychophysical Task

The task was a two-alternative forced-choice orientation discrimination task using procedurally rendered Gabor stimuli. Each trial began with a randomized fixation period of 400–600 ms, followed by a Gabor stimulus oriented at either $+45^\circ$ or -45° for 200 ms. The participant responded using the F/J keys: the F key corresponded to -45° , and the J key corresponded to $+45^\circ$. The application recorded stimulus onset, response key, reaction time, correctness, and task difficulty together with frame-

Table 2 Hardware and software configuration

Item	Reported value
Laptop/PC model	mouse Computer G-Tune P517G60ZO21CNHWT
CPU	Intel Core i7-13620H, 10 cores (P6+E4), 16 threads
GPU	NVIDIA GeForce RTX 5060 Laptop GPU
RAM	32 GB
Storage	1 TB SSD
Operating system	Windows 11 Home 64-bit; exact deployment-time build not logged
Display	15.6-inch display, 2560×1440 pixels, 165 Hz
Browsers used	Mozilla Firefox and Google Chrome
Browser versions	Firefox user-agent was available when recorded in exported metadata; exact Chrome deployment-time version was not systematically logged
Webcam type	Built-in laptop webcam
Webcam model	Exact internal webcam model not logged
Camera capture API	Browser <code>getUserMedia</code> API
Requested camera constraints	Ideal 640×480 pixels, ideal 30 fps
Actual camera resolution	Browser/device-negotiated; not programmatically logged as a verified deployment-time value
Actual camera frame rate	Browser/device-negotiated; not programmatically logged as a verified camera frame rate
Screen brightness	Fixed during the deployment; exact percentage not logged
Auto-exposure	Device/browser default behavior; not manually constrained and exact state not programmatically logged
Auto-white-balance	Device/browser default behavior; not manually constrained and exact state not programmatically logged

Table 3 Deployment environment and lighting conditions

Item	Laboratory	Home
Room type	Indoor university laboratory/research room	Indoor private home room
Time of day	Daytime	Daytime
Primary lighting	General indoor artificial lighting	General indoor artificial lighting
Window presence	Windows present	Windows present
Curtain/blind condition	Curtains present and closed	Curtains present and closed
Natural light contribution	Direct natural light was not used as the primary illumination source	Direct natural light was not used as the primary illumination source
Camera-derived brightness range observed during setup/checks	105–114	102–111
External light-meter measurement	External light-meter values were not used; the reported ranges are camera-derived setup/check information, not lux measurements.	
Screen brightness	Fixed across sessions using the built-in laptop display setting.	
Task display	Stimuli were presented on the built-in laptop display. Screen luminance was part of the tested configuration rather than an independently manipulated factor.	
Window aggregation	Exported window-level features used 10-s windows with 5-s overlap.	
Brightness-change indicator aggregation	The frame-to-frame green-channel brightness-change indicator was summarized in 10-s windows with 5-s overlap.	
Clouds/screen flicker/power cycles	Not treated as observed experimental events; possible brightness variation was handled only as camera-image-derived record variation.	

level sensor-derived features. To maintain adaptive task difficulty, the implementation used a standard 1-up 1-down staircase procedure on stimulus contrast. Stimulus contrast was adjusted by relative $\pm 5\%$ steps according to the previous response. These task parameters are reported for reproducibility; the instruction blocks are not interpreted as validated cognitive or motivational manipulations.

4.7 Dependent Measures and Pre-processing

All exported logs were processed sequentially. The primary behavioral measure was reaction time (RT), with an operational analysis range of 100–1500 ms. The primary sensor-derived measures were rPPG-oriented ROI green-channel values, head-motion estimates, and the frame-to-frame green-

channel brightness-change indicator. Behavioral trials were included in descriptive task analyses when reaction times satisfied $100 \leq RT \leq 1500$ ms. The 80% target referred to the proportion of completed trials whose RTs fell within this study-specific range; it was exploratory and study-specific, not preregistered, independently validated, or intended as a general feasibility threshold. Camerale used MediaPipe FaceMesh to obtain facial landmarks in the browser. Frame records forwarded from the FaceMesh callback were synchronized with task events during active sessions. The feature extractor defines a binary landmark-availability flag, `quality`, where 1 indicates that FaceMesh landmarks were returned and 0 indicates that no landmarks were returned. In the current runtime, however, the camera manager forwards results to the logging callback only when FaceMesh landmarks are returned; therefore, the saved frame records in this deployment represent landmark-returned exported frames. For each exported 10-s window, `mean_quality` was computed as the mean of the `quality` flag over the exported frame records in that window. Thus, `mean_quality` describes landmark availability within exported frame/window records and not face-detection availability across all camera frames. This value is not a continuous MediaPipe confidence score, an indicator of rPPG measurement quality, or evidence of physiological measurement accuracy. The FaceMesh `minDetectionConfidence` and `minTrackingConfidence` parameters were both set to 0.5 as implementation settings. They were not used as analysis thresholds.

5. Results

The results are reported descriptively as sensor-derived observations under the tested configuration. The deployment completed all 10 planned sessions and 600 trials. Of the 600 completed trials, 541 trials (90.2%) had reaction times between 100 and 1500 ms, exceeding the exploratory 80% reference level. Trials within this RT range were 256/300 in the laboratory setting and 285/300 in the home setting. Timestamped task-event and rPPG-sensing records were exported for all 10 sessions in a temporally alignable form, with timestamp fields that allowed task and face-image-derived frame/window records to be related on the same session timeline. Descrip-

tively, all 299 exported windows had `mean_quality` = 1.0; this exported-window summary was not used as the primary evidence for RQ1. These results describe the available analyzed trials and exported windows under the tested configuration and do not demonstrate general feasibility across devices, users, or environments.

5.1 Task and Exported Feature Summaries

Table 4 summarizes task-level and exported feature summaries in the laboratory and home contexts. After RT exclusion, mean RT was 721.1 ± 257.7 ms in the laboratory setting and 676.5 ± 249.5 ms in the home setting. Accuracy was 85.5% and 82.1%, respectively. Window-level exported records showed descriptive differences in ROI green-channel values, motion estimates, and the brightness-change indicator between settings. Because lighting, screen light, posture, camera geometry, task timing, and participant state were not independently controlled, these differences are reported only as deployment-context differences in exported records.

5.2 Task-Event and Sensor-Derived Observations

To address RQ2, we compared descriptive task-synchronized records between the laboratory and home contexts. The exported records showed differences in rPPG-oriented ROI green-channel values, motion estimates, and the frame-to-frame green-channel brightness-change indicator. Because lighting, screen light, posture, camera geometry, task timing, and participant state were not independently controlled, these differences are reported only as deployment-context differences in exported records. They should not be interpreted as evidence that the instruction blocks induced psychological, cognitive, motivational, or physiological effects. Motion estimates are reported as exported face-derived records for descriptive comparison between the laboratory and home contexts. They are not interpreted as evidence of motion handling or correction of rPPG degradation.

The instruction contexts were used only as task-interface contexts in the deployment. We do not report them as evidence that evaluation pressure, reward motivation, cognitive effects, or motivational effects were induced. Descriptive values are reported by instruction context only to document that the log-

Table 4 Descriptive task and exported feature summaries after reaction-time exclusion.

Metric	Laboratory	Home	Notes
Trials within RT range	256/300 (85.3%)	285/300 (95.0%)	Trial-level; RT 100–1500 ms
Mean RT within RT range	721.1 $\pm$ 257.7 ms	676.5 $\pm$ 249.5 ms	Trial-level included trials
Accuracy within RT range	85.5%	82.1%	Trial-level included trials
rPPG-oriented ROI green-channel value	46.69 $\pm$ 6.80	34.32 $\pm$ 2.72	Window-level; windows.csv
Motion estimate	0.0011 $\pm$ 0.0005	0.0010 $\pm$ 0.0004	Window-level; windows.csv
Frame-to-frame green-channel brightness-change indicator	0.0504 $\pm$ 0.0509	0.0707 $\pm$ 0.0337	Window-level; windows.csv

ging pipeline operated across different task-interface contexts. In the laboratory context, the negative instruction block showed a lower mean RT than the neutral block in this dataset. In the home context, the mean RT values across instruction blocks were closer together. Because the deployment used one author-participant and did not isolate instruction effects from environment, lighting, screen light, posture, camera geometry, or measurement-pipeline factors, these RT patterns are reported only as descriptive task-event observations.

5.3 rPPG-Oriented ROI Estimates

The exported data include rPPG-oriented ROI brightness values and related quality indicators, but no ECG or contact PPG reference was collected. Therefore, this paper does not interpret ROI-derived or block-level summaries as validated heart-rate changes. The appropriate interpretation is that the tested deployment yielded face-derived and sensor-derived exported records whose values differed descriptively between the tested laboratory and home contexts.

6. Discussion

The main implication of this deployment is not that Camerale establishes a general design pattern or that home/laboratory differences demonstrate physiological effects. This limited deployment instead identifies practical requirements and safeguards for future edge-side browser-native logging systems: explicit operational criteria, task-event/frame-record timestamp alignment, feature-only local persistence, hardware/environment reporting, and cautious interpretation when raw facial video and reference physiological sensors are not retained. Under the tested configuration, Camerale completed all planned sessions/trials, had 541/600 trials (90.2%) within the study-specific RT range, and exported task-event and rPPG-sensing records in a temporally alignable form for all sessions. The

laboratory/home differences should be interpreted as mixed deployment-context differences rather than psychological, cognitive, motivational, or physiological effects.

Camerale’s useful system property is integration: task timing, frame-level feature extraction, quality indicators, and local export are handled in one client-side workflow. This can reduce raw-video handling and server-side timestamp reconciliation, but it also removes the possibility of visually inspecting many artifacts after data collection. The trade-off makes quality-aware logging important. The relevant indicators include landmark availability, rPPG-oriented ROI values, head-motion estimates, and camera-image-derived brightness-change records; these values can support pre-session checks and descriptive inspection, but they are not proof that rPPG estimates are accurate or that motion-induced rPPG degradation has been corrected.

The laboratory/home differences observed in this deployment should be interpreted as mixed deployment-context differences that may reflect environmental conditions, screen light, posture, camera geometry, the measurement pipeline, task timing, participant state, or their mixture. The study was not designed to isolate psychological or physiological state changes from these factors, and it cannot validate the psychological effectiveness of the negative or positive instruction contexts. Future deployments should therefore incorporate task-event/frame-record timestamp alignment, landmark-availability logging and its limitations, documented hardware/browser/camera settings, explicit reporting of camera constraints and actual camera settings, fixed screen-brightness settings, pre-session camera/lighting checks, landmark-availability warnings, motion and camera-derived brightness-change records, environment reporting, and specified quality-aware analysis rules. If privacy-

sensitive raw video is not stored, these safeguards become more important because many artifacts cannot be retrospectively inspected from images; for studies that aim to make claims about physiological responses, a reference physiological sensor and a multi-participant design are necessary.

6.1 Limitations

This exploratory N=1 author-participant deployment used one primary laptop/camera configuration, no ECG or contact PPG reference, and no comparison with existing experiment runtimes, browser rPPG tools, or cloud APIs. It did not test multiple participants, devices, or home environments, and it did not experimentally separate lighting, screen light, posture, camera placement, task timing, participant state, and task context. The implementation does not correct motion-induced rPPG degradation, and the logs do not store continuous confidence values for all frames; therefore, failed landmark-detection frames and overall tracking-success rates cannot be recovered. RTs were based on browser-side `performance.now()` timestamps; physical stimulus onset was not externally verified. Future work should use reference physiological sensors, controlled posture/camera conditions, predefined artifact-handling rules, and continuous confidence logging with specified operational thresholds. The study does not claim clinical or medical accuracy, general deployability, general feasibility, or validated cognitive, motivational, psychological, or physiological effects; it presents an implementation example of browser-native, task-aligned, client-side feature logging under limited laboratory and home conditions.

7. Conclusion

This paper presented Camerale as an edge-side, browser-native framework for remote psychophysiological experiments with rPPG sensing. In the present single-case author-participant deployment, Camerale completed all 10 planned sessions and 600 trials and exported temporally alignable task-event and rPPG-sensing records under the tested laptop, camera, browser, environment, posture, lighting, and task-execution context. Of the 600 completed trials, 541 trials (90.2%) had reaction times within the study-specific RT range, exceeding the exploratory 80% target. These results provide limited implementation evidence and design considerations for future

Camerale versions, but they do not establish general feasibility, robustness, validity in other settings, validated rPPG accuracy, or psychological/physiological effects.

Declarations

Data availability. The dataset cannot be made publicly available because it contains privacy-sensitive face-derived records from a domestic deployment context.

CRedit authorship contribution statement. **Kotaro Furukawa:** Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing - original draft.

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